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Speaking guide

Directions for delivering `Hunter_SST_techniques.pptx`, 34 slides (31 presented, 3 backup). Not a script. Each entry gives the time to spend, what the slide has to establish, the numbers to say out loud, and what to leave for questions. Total as laid out is ≈22 minutes; the cut-list at the end brings it to 20.

Conventions: `[click]` counts the build steps, so the cue column matches the deck. "Say" means the number should be spoken, not just shown. Deltas are quoted twice, in mIoU points and as a percentage of the arm they were added to, because a point means different things at 55 and at 77; say whichever fits the sentence, and the ledger slide shows both. "Skip unless asked" marks material that is defensible but not load-bearing.

Part I. The baseline and the metric (slides 1 to 4, ≈3:05)

1 · Title (35s). Establish what the project is: a reproduction of Static Segmentation by Tracking, plus an attempt to improve the loss it fine-tunes with. Set the frame for the whole talk, which is a ledger: one baseline, seven techniques, each measured the same way. Flag that the last section changes the model rather than the method, because that is where the ordering reverses.

2 · The task (50s, 6 clicks). Walk the five panels left to right, one click each, naming what each one is. Two things to establish: the method turns segmentation into two-frame tracking, and the render is real project output rather than a diagram. Land on the final click, which is the failure: the query is pinned the other way up and the prediction inherits the reference's layout. Say that four of the seven techniques exist because of this one behaviour.

3 · The metric (45s, 8 clicks). Define the unit before any number is spoken, because every figure after this is in it. Build the five steps, then the worked example, then the semantics card. The line to land is that traits are averaged, not pixels, so a hair-thin antenna counts as much as the elytra. Say 0.655 and point at the zero: that trait was present in the ground truth and never claimed, and it costs the image 0.13 on its own. Mention that a trait predicted but absent from the ground truth contributes no term of its own, and hurts only by taking pixels from traits that are present.

4 · The premise (55s, 4 clicks). Say all four numbers: 55.96 with spread 7.90 against a target of 61.9 ± 3.7, and 68.54 ± 3.15 against 81.0 ± 1.0. Then the protocol line, which is the methodological claim of the talk: twenty seeded draws, the same draws for both arms, paired per seed, a bootstrap interval, and a band recorded before the run. Close on the spread, and give the reason it is treated as a result: it measures sensitivity to which reference was drawn, so a matched mean with double the variance is not a reproduction.

Part II. Support-side techniques (slides 5 to 9, ≈4:10)

5 · Curation (55s, 5 clicks). One click per step. The point to emphasise is the threshold discipline: 55.6 degrees is the training pool's own circular standard deviation, so it is a property of how specimens are pinned rather than a value chosen for its result. Mention that a test pins it. The train-only step is worth ten seconds because a reviewer will otherwise ask about leakage.

6 · The draw (45s, 3 clicks). Frame it as a defect that curation itself introduced. The instructive part is the middle click: stratified random sampling fails here because twenty draws over twelve species is a coupon-collector problem, and three species went unused when measured. Round-robin is the fix. Do not spend time on the seeding details.

7 · What those two bought (55s, 2 clicks). Say 55.96, 58.54, 63.20, which is +7.24 or +12.9%, and the spread sequence 7.90, 5.77, 3.89 against the target's 3.70. Then the per-trait breakdown, which is what makes this mechanistic rather than an average that moved: head 43.6 to 55.9 for curation, pronotum plus 8.7 for balancing. Each fix moved the trait its own diagnostic predicted.

8 · Placement curation (50s, 4 clicks). Establish that butterflies have a different defect from beetles, so the same rule takes a different statistic. Say the result, 68.54 to 70.30, which is +1.90 or +2.8%, and note that the spread change, 3.15 to 1.36, matters more than the mean. Close on the disjointness of the two criteria, which is the evidence that both mechanisms are real.

9 · More references (45s, 3 clicks). Contrast the two datasets: beetle +9.96, which is +15.8%, against butterfly +1.48, which is +2.1%. Explain the asymmetry rather than just reporting it. Averaging over five references fixes defects whose cause is the reference draw, so it fixes beetle and not butterfly. Note the ninefold spread collapse on butterfly as the signature of a defect that is invariant to reference choice. This is the transition into Part III.

Part III. The query-side defect (slides 10 to 11, ≈1:30)

10 · The flip (50s, 3 clicks). Say 37 percent, 74.95 upright, 0.48 inverted, and that the weighted combination reproduces the measured 47.4 to two decimals. The strongest evidence is the second click, the copy baseline scoring zero with no model involved, which makes it a data property. Third click explains why nothing reference-side moved it.

11 · The general form (40s, 2 clicks). One correlation, minus 0.87 over five hundred images, and the quartile spread from 82.8 down to 50.8. Establish two things: the top quartile already exceeds the target, and the test set cannot be curated. Skip unless asked: only 46 percent of the displacement is a global offset, so cropping would not fix it.

Part IV. Orientation TTA (slides 12 to 16, ≈4:15)

This is the centre of the talk. Do not rush it, and do not cut it.

12 · Section title (15s). One sentence of framing: the largest single gain, and the least machinery.

13 · Mechanism, part 1 (60s, 6 clicks). The left column builds the two passes, the right column the three properties. Emphasise that image and mask rotate together, that only the reference rotates, and that the rotated variant needs its own cache key. Mention that the bottom-left panel is the project's own render rotated by the same operation the code applies, so nothing here is an illustration.

14 · Mechanism, part 2 (60s, 5 clicks). Read the criterion off the code block rather than paraphrasing it. Two properties deserve a sentence each: no ground truth is consulted, so the protocol stays one-shot, and caching both orientations keeps the cost at two passes rather than ten. Close on the limitation, which is that it selects between two predictions and cannot improve either.

15 · Results (60s, 2 clicks). Say +5.44 (+7.6%) and +4.88 (+6.9%) on butterfly, then +2.04 (+2.8%) on beetle at five-shot and +0.66 (+1.0%) at one-shot. On the right panel, say 47.40 to 73.27, a gain of +25.87 or +54.6%, against a ceiling of 74.95, which is 94 percent of the deficit. Note that the silhouette score rises with it, so this is re-segmentation rather than better assignment. Second click is the variance cost. Skip unless asked: the rotated-chosen counter postdates this bundle, so the recovery mechanism is consistent with but not proven to be the flipped cohort.

16 · The 2×2 (50s, 3 clicks). Establish the second, reference-side fix and that attention selects rather than blends. Then the square, all against the same 71.78 base: +4.02 (+5.6%), +5.44 (+7.6%), +4.80 together, an interaction of −4.66 or −6.5%. The procedural lesson is worth stating explicitly, that a one-dimensional sweep would have called both additive and shipped the worse arm.

Part V. The loss (slides 17 to 23, ≈5:00)

17 · Section title (15s). Mark the shift: everything before this is inference-time, this is the only technique that changes weights.

18 · The cycle (55s, 6 clicks). Walk the four legs with the panels. Establish the appeal, which is that supervision comes from a mask you already have, so it can train on unlabelled queries.

19 · Two preconditions (55s, 4 clicks). The differentiability fix first: thresholding severs the graph, the reparameterisation reconstructs the same values on a linear path, and the forward-to-reverse ratio goes from 0.00 to 26, and to 351 per-trait. Then the supervision axis, and the law it implies: each objective pays only where it has room, which predicts butterfly gains and beetle does not.

20 · SCNP (60s, 4 clicks). Cover the transform, then why it was a candidate here (the cycle supplies a genuine one-hot target), then both attempts. Say minus 0.09 significant on the merged form, and that the module's own documentation pre-registered beetle as the detrimental case before the run. For the composition, say the prediction was plus 0.5 to plus 2 and the measurement was minus 0.01, inside the recorded band. If time is short, compress the transform and keep the pre-registration.

21 · Every loss variant (55s, 2 clicks). Do not enumerate the arms; the figure does that. Establish the pattern on beetle, name the two arms that hurt badly (full fine-tuning at minus 3.95, the margin fold at minus 3.62), and say why the margin fold failed, which is that training the statistic reshapes the masks TTA chooses between. Note explicitly that this slate is on SAM 2, because Part VI reverses it.

22 · The pot (50s, 2 clicks). The correction to the earlier headroom figure: the quoted number was a correlate, the measured ceiling is 2.65 on beetle and 0.38 on butterfly. Then the sentence the last section depends on, that the remainder is regions the tracker never proposes. Flag it as something to remember.

23 · Scoring (35s, 2 clicks). Three readings of the same predictions: 63.20, 64.69, 65.20. The point is that two points of spread exist with no model change, which is why every fine-tune delta was re-measured under all three. Mention that the model-swap deltas are larger and do move between readings, and that both are quoted. First candidate for cutting.

Part VI. The model swaps and close (slides 24 to 31, ≈7:15)

24 · The ledger (60s, 1 click). Pause and let the audience read before speaking. Every row carries both readings, the mIoU delta and the same gain as a percentage of the arm it was added to; the percentages are what make a +1.59 on beetle and a +1.90 on butterfly comparable. Then give the shape: on SAM 2 everything that paid changed what the model is shown. Point out the two largest negatives as compositions assumed safe, the three new positive rows as the model swap, and the bottom row as the other model's headline feature. Leave this slide up during questions.

25 · Where the numbers land (40s, 2 clicks). Beetle passes both gates, butterfly does not, and roughly five of butterfly's last six points sit below the silhouette ceiling. That is the argument for changing the model rather than adding method.

26 · The two swaps (50s, 4 clicks). Introduce both families and what was held fixed. The cross-family pairing discipline is worth one sentence for this audience: the draws are model-independent and hash-pinned, the registry refuses such comparisons by default, and this stage uses one whitelisted relaxation that records what it ignored. Then the new arm, LoRA on the memory attention.

27 · SeC-4B (70s, 3 clicks). Read the deployment row first: −2.79 on butterfly and −3.02 on beetle, which is −3.6% and −4.7%. Note its oracle pot is in the same range, so the proposal boundary did not move. Then the concept injection, −5.92 and −6.23, which is −8.6% and −10.0%, with 0 and 1 of 20 seeds improving. Report the falsified pre-registration directly: the native gate was predicted quiet, and it fires on 0.585 of butterfly episodes and 0.869 of beetle ones, so the gate-off cells are the charitable reading. Close on the one positive, the loss stack at plus 0.60 and plus 0.58.

28 · SAM 3 (75s, 2 clicks). Four rungs, every one significant: +2.14 (+3.4%), +0.99 (+1.5%), +1.59 (+2.4%), +1.96 (+2.9%). Then the headline, 64.12 to 69.88, which is +5.77 or +9.0% against the SAM 2 stack, and 70.56 under the paper's own full metric. Second click is butterfly as the control, flat at 76.5 to 76.6, which is the third independent confirmation that its residual is protocol rather than capacity.

29 · Why it worked (65s, 3 clicks). This is the analytical payoff, so give it room. SAM 3's own assignment pot is smaller than SAM 2's, 1.80 against 2.65, yet the stack and the arch arm delivered 3.55 points, or +5.4% on the arm they were added to, which is only possible if the ceiling moved. It did, 66.76 to 71.08. Connect it back to slide 21 explicitly. Add that the loss arms were written on per-object logit stacks so they could transfer, which is why the port was cheap.

30 · What is next (55s, 3 clicks). Present the projection as a projection. The bands come from measured components, two independent caps agree near 81, and there is a pre-registered disqualifier at 72 for the five-shot base. Mention the conditioning-frame cap of four as a known trap that is configuration rather than weights. Second candidate for cutting; it is the most likely thing to be asked about anyway.

31 · Close (45s, 4 clicks). One line per click. The through-line to land is that the ordering reversed: on SAM 2 the objective was the wrong place to spend effort, on SAM 3 the objective plus the proposal-forming arm is most of the gain, and butterfly's residual survived both swaps.

Backup slides (32 to 34)

Not presented. They exist for the implementation questions, which is what got asked last time. Jump to them rather than describing the code from memory.

- **32, canonicalization geometry.** The three criteria as computed: the directed head-to-tail centroid vector, the centroid displacement against a per-trait median reference, and completeness, plus the 180 degree test used to build strata.
- **33, pool to draw.** `canonical_pool` to `draw_supports` to the seeding, with the round-robin rule, the orientation-diverse proportion, and the support-set hash.
- **34, what guards it.** The tests that pin the threshold, the leakage check, the dispatch check, and the list of fields recorded in every bundle.

`TECHNICAL_QA.md` is the long form, organised by question. Sections 1 to 3 back slide 32, 4 to 8 back slide 33, 9 to 12 back slide 34. Its last section is a set of one-line answers for questions asked in passing.

Two places in the main talk are worth an explicit offer to go deeper, because they are where the question usually comes from: slide 5 (the threshold, where "is that tuned?" is the natural follow-up) and slide 8 (placement, which is the least familiar of the three criteria to an audience).

Cutting to 20 minutes

In order:

1. **Slide 23**, scoring, 35s. The point survives as one sentence on slide 24.
2. **Slide 30**, the projection, 55s. Keep it as a backup for questions.
3. **Slide 11**, displacement quartiles, 40s, if slide 10 landed clearly.
4. **Slide 6**, the round-robin, 45s. Mention it in one line on slide 5.

Do not cut 3, 13 to 15, 20, or 28 to 29.

Questions to expect

Why does the loss work on SAM 3 when it was null on SAM 2? The binding constraint changed. On SAM 2 every loss arm was bounded by a 2.65-point assignment pot, about 4% of the score. The arch arm is not an assignment method: it trains the module that forms proposals, and the oracle ceiling moved 4.3 points, which is the direct evidence.

Is the cross-family comparison legitimate? The support draws are model-independent by construction and hash-pinned, so both families saw the same twenty references. The registry refuses cross-family pairings by default; this stage used one whitelisted relaxation, and every row records the fields ignored and both their values.

Why is the confidence criterion trustworthy? It is not, entirely. That is the variance cost on slide 15, and on one family and dataset combination it goes negative, SeC on beetle at minus 1.80. It is right often enough to be worth 5.4 points net on butterfly. Improving the criterion is open headroom.

Is curating the reference pool consistent with a one-shot protocol? Curation reads training masks only, never test. The uncomfortable part is the accounting: the best butterfly arm reads about 1,700 labelled masks to choose one reference. The defensible version is that curation stands in for human judgement, and a human does not need 1,700 masks. It is testable by shrinking the pool until the drawn set stops changing.

Could the SAM 2 nulls have been an implementation fault? The gradient measurement on slide 19, the training diagnostics, and an exact checkpoint round-trip all say the machinery works, and the same machinery produces significant positives on SAM 3.